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Effects of source of trace minerals on the uterine metabolome profile in dairy cows

Objectives were to evaluate the effects of replacing supplemental trace minerals as sulfate (**STM**) with hydroxychloride (**HTM**) sources of Cu, Mn, and Zn from 246 d of gestation to 105 d in milk (**DIM**) on day 16 uterine metabolome profile and early pregnancy outcomes. Holstein cows (n=141) were assigned in a randomized complete block design to STM or HTM treatments. Prepartum diets contained 15, 58, and 66 mg/kg of diet DM as Cu, Mn, and Zn, respectively, whereas postpartum diets contained 19, 65, and 77 mg/kg diet DM of the same trace minerals. Cows had the estrous cycle synchronized and 84 received artificial insemination (**AI**) at 54 DIM, whereas 57 remained non-inseminated. On day 16 of the estrous cycle, the uterine horn ipsilateral to corpus luteum was sampled using a cytobrush for metabolomic analysis. Inseminated cows were flushed for conceptus recovery. Cows with conceptus or interferon tau (**IFNT**) >200 pg/mL were considered pregnant. Cytobrush samples from 41 pregnant (20 STM, 21 HTM) and 32 non-inseminated (16 STM, 16 HTM) were analyzed using the MxP Quant500 kit with 624 metabolites including primarily lipids and amino acids. Data were analyzed by linear mixed-effects models using the MIXED or GLIMMIX procedures of SAS/STAT, whereas metabolomic data were analyzed in MetaboAnalyst 6.0 with adjustments for parity and sampling cohort. Replacing STM with HTM did not affect pregnancy per AI, conceptus length, or IFNT concentration in the uterine fluid. After quality control, metabolomic analysis quantified 343 of the 624 metabolites investigated. Treatment did not affect the concentration of any of the 343 metabolites quantified after adjusting for false discovery, and this response was irrespective of pregnancy status. These data suggest that source of trace minerals did not influence conceptus characteristics or the composition of the uterine lumen fluid on day 16 of the estrous cycle or pregnancy.

Item	STM	HTM	SEM	*P*-value
Pregnant day 16, %				
All cows	56.2	67.7	7.7	0.29
Synchronized cows	63.6	76.6	7.6	0.23
Conceptus length, cm	8.17	7.60	1.14	0.70
Flush IFNT, ng/mL	11.6	17.6	6.3	0.47

cow, uterine metabolome, trace mineral